

MATH1014 Calculus II Tutorial

Week 11

Wenkai Fan

1 Review

1.1 Absolute and Conditional Convergence

Given any series $\sum a_n$, we can consider the corresponding series of absolute values $\sum |a_n|$.

- **Absolute Convergence:** A series $\sum a_n$ is called *absolutely convergent* if the series of absolute values $\sum |a_n|$ is convergent.
- **Theorem:** If a series $\sum a_n$ is absolutely convergent, then it is convergent. (Absolute convergence implies convergence).
- **Conditional Convergence:** A series $\sum a_n$ is called *conditionally convergent* if it is convergent but not absolutely convergent. (That is, $\sum a_n$ converges, but $\sum |a_n|$ diverges).

1.2 The Alternating Series Test (AST)

An alternating series is a series whose terms are alternately positive and negative, e.g., $\sum_{n=1}^{\infty} (-1)^{n-1} b_n$ or $\sum_{n=1}^{\infty} (-1)^n b_n$, where $b_n > 0$. If the alternating series $\sum_{n=1}^{\infty} (-1)^{n-1} b_n$ satisfies:

1. $b_{n+1} \leq b_n$ for all n (the sequence b_n is decreasing), and
2. $\lim_{n \rightarrow \infty} b_n = 0$,

then the series is **convergent**. **Note:** To show that b_n is decreasing, we can often consider the related function $f(x)$ where $f(n) = b_n$, and show that its derivative $f'(x) \leq 0$ for large x .

1.3 The Ratio Test

The Ratio Test is particularly useful for series involving factorials (e.g., $n!$) or exponentials (e.g., 2^n). Let $L = \lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right|$.

- If $L < 1$, then the series $\sum a_n$ is **absolutely convergent** (and therefore convergent).
- If $L > 1$ or $\lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| = \infty$, then the series $\sum a_n$ is **divergent**.
- If $L = 1$, the Ratio Test is **inconclusive**; no conclusion can be drawn. We must use another test.

1.4 The Root Test

The Root Test is particularly useful when the entire n -th term is raised to the power of n . Let $L = \lim_{n \rightarrow \infty} \sqrt[n]{|a_n|} = \lim_{n \rightarrow \infty} |a_n|^{1/n}$.

- If $L < 1$, then the series $\sum a_n$ is **absolutely convergent** (and therefore convergent).
- If $L > 1$ or $\lim_{n \rightarrow \infty} \sqrt[n]{|a_n|} = \infty$, then the series $\sum a_n$ is **divergent**.
- If $L = 1$, the Root Test is **inconclusive**; we must use another test.

2 Exercises

2.1 Alternating Series & Absolute/Conditional Convergence

Exercise 2.1. Check whether the following series are absolutely convergent, conditionally convergent or divergent.

1. $\sum_{n=1}^{\infty} (-1)^{n+1} \sin\left(\frac{\pi}{2n}\right)$
2. $\sum_{n=1}^{\infty} (-1)^n \frac{\ln^2 n}{n}$
3. $\sum_{n=1}^{\infty} (-1)^{n+1} \frac{\ln(2 + \frac{1}{n})}{\sqrt{(3n-2)(3n+2)}}$
4. $\sum_{n=1}^{\infty} \frac{1}{\sqrt{n}} \cos\left(\frac{n\pi}{3}\right)$

Solution. **(1)** Let $a_n = (-1)^{n+1} \sin\left(\frac{\pi}{2n}\right)$. First, we check for absolute convergence by considering the series $\sum |a_n| = \sum \sin\left(\frac{\pi}{2n}\right)$. Note that for $n \geq 1$, $0 < \frac{\pi}{2n} \leq \frac{\pi}{2}$, so $\sin\left(\frac{\pi}{2n}\right) > 0$. We use the Limit Comparison Test with $b_n = \frac{\pi}{2n}$:

$$\lim_{n \rightarrow \infty} \frac{|a_n|}{b_n} = \lim_{n \rightarrow \infty} \frac{\sin\left(\frac{\pi}{2n}\right)}{\frac{\pi}{2n}} = 1 > 0.$$

Since $\sum b_n = \frac{\pi}{2} \sum \frac{1}{n}$ is a divergent harmonic series, $\sum |a_n|$ is **divergent** by the Limit Comparison Test. Thus, the series is not absolutely convergent.

Now, we check for conditional convergence using the Alternating Series Test. Let $b_n = \sin\left(\frac{\pi}{2n}\right)$.

- As n increases, $\frac{\pi}{2n}$ decreases. Since $f(x) = \sin x$ is increasing on $(0, \frac{\pi}{2}]$, b_n is a decreasing sequence. So $b_{n+1} \leq b_n$.
- $\lim_{n \rightarrow \infty} b_n = \lim_{n \rightarrow \infty} \sin\left(\frac{\pi}{2n}\right) = \sin(0) = 0$.

Since both conditions are met, the series $\sum a_n$ is convergent by the Alternating Series Test. **Conclusion:** The series is **conditionally convergent**.

(2) Let $a_n = (-1)^n \frac{\ln^2 n}{n}$. First, check for absolute convergence: $\sum |a_n| = \sum \frac{\ln^2 n}{n}$. We compare it with the divergent harmonic series $\sum \frac{1}{n}$.

$$\lim_{n \rightarrow \infty} \frac{\frac{\ln^2 n}{n}}{\frac{1}{n}} = \lim_{n \rightarrow \infty} \ln^2 n = \infty.$$

Since $\frac{\ln^2 n}{n} \geq \frac{1}{n}$ for $n \geq 3$, by the Comparison Test, $\sum |a_n|$ is **divergent**. The series is not absolutely convergent.

Next, we check for conditional convergence using the Alternating Series Test. Let $b_n = \frac{\ln^2 n}{n}$.

- To show b_n is decreasing, we consider $f(x) = \frac{\ln^2 x}{x}$ and find its derivative:

$$f'(x) = \frac{2 \ln x \cdot \frac{1}{x} \cdot x - \ln^2 x \cdot 1}{x^2} = \frac{2 \ln x - \ln^2 x}{x^2} = \frac{\ln x(2 - \ln x)}{x^2}.$$

For $x > e^2 \approx 7.389$, $\ln x > 2$, so $2 - \ln x < 0$. Thus $f'(x) < 0$. This implies that b_n is decreasing for $n \geq 8$.

- We check the limit: $\lim_{n \rightarrow \infty} \frac{\ln^2 n}{n}$. By L'Hôpital's Rule (applied twice):

$$\lim_{x \rightarrow \infty} \frac{\ln^2 x}{x} = \lim_{x \rightarrow \infty} \frac{2 \ln x \cdot \frac{1}{x}}{1} = \lim_{x \rightarrow \infty} \frac{2 \ln x}{x} = \lim_{x \rightarrow \infty} \frac{2/x}{1} = 0.$$

Since b_n is eventually decreasing and its limit is 0, the series converges by the AST. **Conclusion:** The series is **conditionally convergent**.

(3) Let $a_n = (-1)^{n+1} \frac{\ln(2 + \frac{1}{n})}{\sqrt{(3n-2)(3n+2)}}$. First, check for absolute convergence: $\sum |a_n| = \sum \frac{\ln(2 + \frac{1}{n})}{\sqrt{9n^2 - 4}}$. For large n , $\ln(2 + \frac{1}{n}) \approx \ln 2$, and $\sqrt{9n^2 - 4} \approx 3n$. So we use the Limit Comparison Test with $b_n = \frac{1}{n}$:

$$\lim_{n \rightarrow \infty} \frac{|a_n|}{b_n} = \lim_{n \rightarrow \infty} \frac{n \ln(2 + \frac{1}{n})}{\sqrt{9n^2 - 4}} = \lim_{n \rightarrow \infty} \frac{\ln(2 + \frac{1}{n})}{\sqrt{9 - \frac{4}{n^2}}} = \frac{\ln 2}{3} > 0.$$

Since $\sum \frac{1}{n}$ is divergent, $\sum |a_n|$ is **divergent**. The series is not absolutely convergent.

Now, use the Alternating Series Test. Let $b_n = \frac{\ln(2 + \frac{1}{n})}{\sqrt{(3n-2)(3n+2)}}$.

- As n increases, $2 + \frac{1}{n}$ decreases, so the numerator $\ln(2 + \frac{1}{n})$ is positive and decreasing.
- As n increases, the denominator $\sqrt{9n^2 - 4}$ is positive and increasing.
- A decreasing positive numerator divided by an increasing positive denominator results in a decreasing sequence. Thus $b_{n+1} \leq b_n$.
- $\lim_{n \rightarrow \infty} b_n = \lim_{n \rightarrow \infty} \frac{\ln(2 + \frac{1}{n})}{\sqrt{9n^2 - 4}} = \frac{\ln 2}{\infty} = 0$.

By the AST, the series converges. **Conclusion:** The series is **conditionally convergent**.

(4) Let $a_n = \frac{1}{\sqrt{n}} \cos\left(\frac{n\pi}{3}\right)$. This is an alternating-like series. Let's list the values of $\cos\left(\frac{n\pi}{3}\right)$ for $n = 1, 2, 3, 4, 5, 6$:

$$\frac{1}{2}, \quad -\frac{1}{2}, \quad -1, \quad -\frac{1}{2}, \quad \frac{1}{2}, \quad 1.$$

The sequence has a period of 6. Notice that the absolute value of a_n is $|a_n| = \frac{1}{\sqrt{n}} \left| \cos\left(\frac{n\pi}{3}\right) \right|$. Since $\left| \cos\left(\frac{n\pi}{3}\right) \right| \geq \frac{1}{2}$ for all n , we have $|a_n| \geq \frac{1}{2\sqrt{n}}$. Since $\sum \frac{1}{2\sqrt{n}}$ is a divergent p -series ($p = \frac{1}{2} < 1$), by the Comparison Test, $\sum |a_n|$ is **divergent**.

To check for conditional convergence, we group the terms according to the value of the cosine. Because the sequence of cosines has period 6 and sums to 0 over a period, we can split the series

into three strictly alternating series:

$$\begin{aligned}\sum_{n=1}^{\infty} a_n &= \sum_{k=1}^{\infty} a_{3k-2} + \sum_{k=1}^{\infty} a_{3k-1} + \sum_{k=1}^{\infty} a_{3k} \\ &= \sum_{k=1}^{\infty} \frac{(-1)^{k-1}}{2\sqrt{3k-2}} + \sum_{k=1}^{\infty} \frac{(-1)^{k-1}}{2\sqrt{3k-1}} + \sum_{k=1}^{\infty} \frac{(-1)^k}{\sqrt{3k}}.\end{aligned}$$

Each of these three series satisfies the conditions of the Alternating Series Test: their absolute terms are decreasing and limit to 0. Thus, all three sub-series are convergent. The sum of convergent series is convergent. **Conclusion:** The series is **conditionally convergent**. $\square$

2.2 The Ratio Test

Exercise 2.2. Check whether the following series are absolutely convergent, conditionally convergent or divergent.

1. $\sum_{n=1}^{\infty} n^2 2^{-n}$
2. $\sum_{n=1}^{\infty} \frac{2^n \cdot n!}{n^n}$
3. $\sum_{n=1}^{\infty} \frac{a^n}{(1+a)(1+a^2)\cdots(1+a^n)} \quad (a > 0)$

Solution. (1) Let $a_n = n^2 2^{-n} = \frac{n^2}{2^n}$. We apply the Ratio Test:

$$L = \lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| = \lim_{n \rightarrow \infty} \frac{(n+1)^2}{2^{n+1}} \cdot \frac{2^n}{n^2} = \lim_{n \rightarrow \infty} \left(\frac{n+1}{n} \right)^2 \cdot \frac{2^n}{2^{n+1}}.$$

Simplifying the expression:

$$L = \lim_{n \rightarrow \infty} \left(1 + \frac{1}{n} \right)^2 \cdot \frac{1}{2} = (1)^2 \cdot \frac{1}{2} = \frac{1}{2}.$$

Since $L = \frac{1}{2} < 1$, by the Ratio Test, the series is **absolutely convergent**.

(2) Let $a_n = \frac{2^n \cdot n!}{n^n}$. We apply the Ratio Test:

$$\begin{aligned}L &= \lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| = \lim_{n \rightarrow \infty} \frac{2^{n+1} \cdot (n+1)!}{(n+1)^{n+1}} \cdot \frac{n^n}{2^n \cdot n!} \\ &= \lim_{n \rightarrow \infty} \left(\frac{2^{n+1}}{2^n} \right) \cdot \left(\frac{(n+1)!}{n!} \right) \cdot \left(\frac{n^n}{(n+1)^{n+1}} \right) \\ &= \lim_{n \rightarrow \infty} 2 \cdot (n+1) \cdot \frac{n^n}{(n+1)^n (n+1)} \\ &= \lim_{n \rightarrow \infty} 2 \cdot \left(\frac{n}{n+1} \right)^n = \lim_{n \rightarrow \infty} \frac{2}{\left(\frac{n+1}{n} \right)^n} \\ &= \lim_{n \rightarrow \infty} \frac{2}{\left(1 + \frac{1}{n} \right)^n}.\end{aligned}$$

Recall the standard limit $\lim_{n \rightarrow \infty} \left(1 + \frac{1}{n}\right)^n = e$. Therefore,

$$L = \frac{2}{e}.$$

Since $e \approx 2.718$, we have $L = \frac{2}{e} < 1$. By the Ratio Test, the series is **absolutely convergent**.

(3) Let $a_n = \frac{a^n}{(1+a)(1+a^2)\cdots(1+a^n)}$. We apply the Ratio Test:

$$L = \lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| = \lim_{n \rightarrow \infty} \frac{a^{n+1}}{(1+a) \cdots (1+a^n)(1+a^{n+1})} \cdot \frac{(1+a) \cdots (1+a^n)}{a^n}.$$

Most terms cancel out, leaving:

$$L = \lim_{n \rightarrow \infty} \frac{a}{1+a^{n+1}}.$$

We must discuss this limit based on the value of $a > 0$:

- **If $0 < a < 1$:** Then $a^{n+1} \rightarrow 0$ as $n \rightarrow \infty$. Thus, $L = \frac{a}{1+0} = a < 1$.
- **If $a = 1$:** Then $a^{n+1} = 1$ for all n . Thus, $L = \frac{1}{1+1} = \frac{1}{2} < 1$.
- **If $a > 1$:** Then $a^{n+1} \rightarrow \infty$ as $n \rightarrow \infty$. Thus, $L = \frac{a}{\infty} = 0 < 1$.

In all cases, $L < 1$. By the Ratio Test, the series is **absolutely convergent** for all $a > 0$. $\square$

2.3 The Root Test & Parameter Discussion

Exercise 2.3. Check whether the following series are absolutely convergent, conditionally convergent or divergent.

1. $\sum_{n=1}^{\infty} (-1)^{n+1} \frac{n^2 x^n}{2^n}$
2. $\sum_{n=1}^{\infty} (-1)^{n+1} \frac{4^n \sin^{2n} x}{n}$

Solution. (1) Let $a_n = (-1)^{n+1} \frac{n^2 x^n}{2^n}$. Because of the n -th powers, we use the **Root Test**:

$$L = \lim_{n \rightarrow \infty} \sqrt[n]{|a_n|} = \lim_{n \rightarrow \infty} \sqrt[n]{\left| \frac{n^2 x^n}{2^n} \right|} = \lim_{n \rightarrow \infty} \frac{\sqrt[n]{n^2} |x|}{2} = \lim_{n \rightarrow \infty} \frac{(\sqrt[n]{n})^2 |x|}{2}.$$

Since $\lim_{n \rightarrow \infty} \sqrt[n]{n} = 1$, we have:

$$L = \frac{1^2 \cdot |x|}{2} = \frac{|x|}{2}.$$

By the Root Test:

- If $L = \frac{|x|}{2} < 1 \implies |x| < 2$, the series is **absolutely convergent**.
- If $L = \frac{|x|}{2} > 1 \implies |x| > 2$, the series is **divergent**.
- If $L = 1 \implies |x| = 2$, the Root Test is inconclusive. We must check the endpoints $x = 2$ and $x = -2$ manually.

Checking boundaries:

When $x = 2$: $a_n = (-1)^{n+1} \frac{n^2 2^n}{2^n} = (-1)^{n+1} n^2$. As $n \rightarrow \infty$, $\lim a_n$ does not exist. By the Test for Divergence, the series diverges.

When $x = -2$: $a_n = (-1)^{n+1} \frac{n^2 (-2)^n}{2^n} = (-1)^{n+1} n^2 (-1)^n = (-1)^{2n+1} n^2 = -n^2$. As $n \rightarrow \infty$, $\lim a_n = -\infty \neq 0$. The series diverges.

Conclusion: The series is **absolutely convergent** for $|x| < 2$, and **divergent** for $|x| \geq 2$.

(2) Let $a_n = (-1)^{n+1} \frac{4^n \sin^{2n} x}{n}$. We use the **Root Test**:

$$L = \lim_{n \rightarrow \infty} \sqrt[n]{|a_n|} = \lim_{n \rightarrow \infty} \sqrt[n]{\frac{4^n |\sin x|^{2n}}{n}} = \lim_{n \rightarrow \infty} \frac{4 |\sin x|^2}{\sqrt[n]{n}}.$$

Since $\lim_{n \rightarrow \infty} \sqrt[n]{n} = 1$, we have:

$$L = 4 \sin^2 x.$$

By the Root Test:

- If $4 \sin^2 x < 1 \implies \sin^2 x < \frac{1}{4} \implies |\sin x| < \frac{1}{2}$. This occurs when $x \in (k\pi - \frac{\pi}{6}, k\pi + \frac{\pi}{6})$ for any integer k . In these intervals, the series is **absolutely convergent**.
- If $4 \sin^2 x > 1 \implies |\sin x| > \frac{1}{2}$. This occurs when $x \in (k\pi + \frac{\pi}{6}, k\pi + \frac{5\pi}{6})$ for any integer k . In these intervals, the series is **divergent**.
- If $4 \sin^2 x = 1 \implies |\sin x| = \frac{1}{2}$. This occurs at the boundary points $x = k\pi \pm \frac{\pi}{6}$. The Root Test is inconclusive, so we check these points manually.

Checking boundaries:

When $|\sin x| = \frac{1}{2}$, we have $\sin^2 x = \frac{1}{4}$. Substituting this back into the original series gives:

$$a_n = (-1)^{n+1} \frac{4^n \left(\frac{1}{4}\right)^n}{n} = \frac{(-1)^{n+1}}{n}.$$

The series becomes $\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n}$, which is the alternating harmonic series. We know that the alternating harmonic series converges (by the Alternating Series Test) but does not converge absolutely (the harmonic series $\sum \frac{1}{n}$ diverges). Thus, it is conditionally convergent.

Conclusion: The series is **absolutely convergent** when $x \in (k\pi - \frac{\pi}{6}, k\pi + \frac{\pi}{6})$.

It is **conditionally convergent** when $x = k\pi \pm \frac{\pi}{6}$.

It is **divergent** when $x \in (k\pi + \frac{\pi}{6}, k\pi + \frac{5\pi}{6})$. □